Chapter 5: Interaction

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Chapter 4 introduced effect modification: the effect of a single treatment A varying across levels of another variable V. Many causal questions, however, concern two or more treatments applied together. If the causal effect of one treatment depends on the value we set for the other, the two treatments **interact**.

This chapter defines interaction between two treatments in two frameworks:

- the counterfactual (potential outcomes) framework, which we have used so far;
- the sufficient-component-cause framework, which describes causal mechanisms.

This chapter is based on Hernán and Robins (2020, chap. 5, pp. 61-74).

The book motivates the chapter with the "looking up" experiment from Chapter 1: suppose we randomize not only whether you look up at the sky but also whether you stand in the street dressed or naked. If the effect of looking up differs between the dressed and naked conditions, the two

treatments interact. When joint interventions are feasible, knowing about interaction tells us which combination of interventions works best.

1 5.1 Interaction Requires a Joint Intervention (pp. 61-62)

1.1 Joint Interventions

Extend the heart transplant example with a second treatment:

- A: heart transplant ($A = 1$) or not ($A = 0$);
- E: multivitamin complex ($E = 1$) or no vitamins ($E = 0$).

Each individual now has four counterfactual outcomes, $Y^{a=1,e=1}$, $Y^{a=1,e=0}$, $Y^{a=0,e=1}$, and $Y^{a=0,e=0}$.

> **Definition 1.1** (Joint Intervention and Joint Counterfactual)**.** A **joint intervention** sets two or more treatments at once. The counterfactual outcome $Y^{a,e}$ is the outcome that would have been observed had we intervened to set A to a and E to e.

Recursive substitution (Hernán and Robins 2020, 61, margin note). The counterfactual Y^a under an intervention on A alone is the joint counterfactual $Y^{a,e}$ evaluated at the value e that E actually takes: $Y^a = Y^{a,E}$. Consistency is a special case of the same substitution: the observed outcome is $Y = Y^A = Y^{A,E}$.

1.2 Interaction in the Counterfactual Framework

> **Definition 1.2** (Interaction on the Additive Scale)**.** There is **interaction between A and E on the additive scale** in the population if the causal risk difference for A when everybody receives E differs from the causal risk difference for A when nobody receives E:
>
> $$\Pr[Y^{a=1,e=1} = 1] - \Pr[Y^{a=0,e=1} = 1] \neq \Pr[Y^{a=1,e=0} = 1] - \Pr[Y^{a=0,e=0} = 1].$$

> **Example 1.1** (Transplant and Vitamins (Hernán and Robins 2020, 61–62))**.** Suppose the causal risk difference for transplant is
> - 0.1 when everybody receives vitamins: $\Pr[Y^{a=1,e=1} = 1] - \Pr[Y^{a=0,e=1} = 1] = 0.1$;
> - 0.2 when nobody receives vitamins: $\Pr[Y^{a=1,e=0} = 1] - \Pr[Y^{a=0,e=0} = 1] = 0.2$.
>
> Because $0.1 \neq 0.2$, transplant and vitamins interact on the additive scale.

1.3 A and E Have Equal Status

Write $p_{ae} \overset{\text{def}}{=} \Pr[Y^{a,e} = 1]$. Moving terms across the inequality in Definition 1.2 shows that the same condition can be stated with the roles of A and E swapped:

$$\begin{aligned} & p_{11} - p_{01} \neq p_{10} - p_{00} & \\ p_{11} - p_{01} - p_{10} + p_{00} & \neq 0 & \text{(subtract } p_{10} - p_{00} \text{ from both sides)} \\ & p_{11} - p_{10} \neq p_{01} - p_{00} & \text{(add } p_{01} - p_{00} \text{ to both sides)} \end{aligned}$$

The last line says that the causal risk difference for vitamins E differs between "everybody transplanted" and "nobody transplanted".

In Example 1.1, the middle line gives $p_{11} - p_{01} - p_{10} + p_{00} = 0.1 - 0.2 = -0.1$, so the last line gives $p_{11} - p_{10} = (p_{01} - p_{00}) - 0.1$: the risk difference for vitamins is 0.1 lower among the transplanted

than among the untransplanted, which is the book's statement that the inequality for A implies the corresponding inequality for E (Hernán and Robins 2020, 62).

Technical Point 5.1: Additive and Multiplicative Scales

No interaction on the additive scale means the joint effect is the sum of the two separate effects:

$$p_{11} - p_{00} = (p_{10} - p_{00}) + (p_{01} - p_{00}).$$

When the two sides differ, the interaction is

- **superadditive** if the left side is greater ($>$);
- **subadditive** if the left side is smaller ($<$).

On the **multiplicative scale** (causal risk ratios), A and E interact if

$$\frac{p_{11}}{p_{00}} \neq \frac{p_{10}}{p_{00}} \times \frac{p_{01}}{p_{00}},$$

with **supermultiplicative** ($>$) and **submultiplicative** ($<$) interaction defined analogously. The additive form follows from the definition in two steps (Hernán and Robins 2020, Technical Point 5.1, p. 63). No interaction means $p_{11} - p_{01} = p_{10} - p_{00}$. Adding p_{01} to both sides gives $p_{11} = p_{10} - p_{00} + p_{01}$. Subtracting p_{00} from both sides gives $p_{11} - p_{00} = (p_{10} - p_{00}) + (p_{01} - p_{00})$, the displayed equality.

Subtracting $(p_{10} - p_{00}) + (p_{01} - p_{00})$ from $p_{11} - p_{00}$ leaves $p_{11} - p_{10} - p_{01} + p_{00}$, so in Example 1.1 (where that quantity is -0.1) the interaction is subadditive.

1.4 Interaction versus Effect Modification

	Effect modification by V	Interaction between A and E
Counterfactuals involved	Y^a	$Y^{a,e}$
Interventions	on A only	joint, on A and E
Status of the two variables	unequal: V is not intervened on	equal

Effect modification is about the causal effect of A only. In Chapter 4, sex modified the effect of transplant, but the effect of sex on death was never considered, so V and A were not on an equal footing. Interaction concerns the joint causal effect of A and E, and its two equivalent definitions treat A and E symmetrically (Hernán and Robins 2020, 62).

2 5.2 Identifying Interaction (pp. 62-64)

Interaction concerns a joint effect, so identifying it requires exchangeability, positivity, and consistency for **both** treatments.

2.1 When E Is Randomized, Interaction Is Effect Modification by E

If vitamins E are randomly and unconditionally assigned, then for each a and e

$$\begin{aligned}
\Pr[Y^{a,e} = 1] &= \Pr[Y^{a,e} = 1 \mid E = e] && \text{(exchangeability for } E\text{: } Y^{a,e} \perp\!\!\!\perp E) \\
&= \Pr[Y^{a,E} = 1 \mid E = e] && (Y^{a,e} = Y^{a,E} \text{ when } E = e) \\
&= \Pr[Y^{a} = 1 \mid E = e] && \text{(recursive substitution, } Y^{a} = Y^{a,E}).
\end{aligned}$$

Substituting into Definition 1.2, interaction on the additive scale becomes

$$\Pr[Y^{a=1} = 1 \mid E = 1] - \Pr[Y^{a=0} = 1 \mid E = 1] \neq \Pr[Y^{a=1} = 1 \mid E = 0] - \Pr[Y^{a=0} = 1 \mid E = 0],$$

which is the definition of additive effect modification of A by E.

So when E is randomized, interaction and effect modification coincide, and the Chapter 4 methods for detecting effect modification by V apply after replacing V by E (Hernán and Robins 2020, 62–63).

2.2 When E Is Not Randomized

We still need the four marginal risks $\Pr[Y^{a,e} = 1]$. Under the usual identifying assumptions for **both** treatments, they can be computed by standardization or IP weighting over the measured covariates.

Equivalently, treat AE as **one treatment with four levels** (11, 01, 10, 00). Identifying interaction is then the familiar problem of identifying the effect of a single treatment, with more treatment values and more counterfactual outcomes.

Hernán and Robins (2020, 64) states that exchangeability, positivity, and consistency for the joint treatment (A, E) are a **sufficient** condition for identifying interaction. Part III of the book shows the condition is not necessary when the two treatments occur at different times. After this chapter, most of Parts I and II return to a single treatment A.

2.3 Effect Modification Without Interaction

If we are willing to assume exchangeability for A but not for E (e.g., A is randomized and we look at subgroups defined by E), we can assess **effect modification by** E but generally not **interaction between** A **and** E: computing the effect of A within strata of E requires no assumptions about the effect of E.

- Chapter 4 wrote such variables as V (no identifying assumptions made for them).
- Nationality V was a **surrogate** effect modifier: it does not act on Y, so it does not interact with A ("no action, no interaction").
- V still modifies the effect of A because it is correlated with a variable that does act on Y and does interact with A.

So there can be effect modification of A by a variable without interaction between A and that variable. The reverse, interaction between A and E without modification of the effect of A by E, is logically possible but probably rare, because it requires dual effects of A and exact cancellations (Hernán and Robins 2020, 64, margin note citing VanderWeele 2009b).

The results so far do not require deterministic counterfactuals. Section 5.3 does: from there on the chapter assumes deterministic counterfactuals and dichotomous treatments and outcome (Hernán and Robins 2020, 64).

3 5.3 Counterfactual Response Types and Interaction (pp. 64-66)

3.1 One Treatment: Four Response Types

Table 1: Response types for one dichotomous treatment (Hernán and Robins 2020, Table 5.1, p. 64)

Type	$Y^{a=0}$	$Y^{a=1}$	Zeus's family (Table 1.1)
Doomed	1	1	Artemis, Athena, Persephone, Ares
Helped	1	0	Hebe, Kronos, Poseidon, Apollo, Hermes, Dionysus
Hurt	0	1	Rheia, Leto, Aphrodite, Zeus, Hephaestus, Polyphemus
Immune	0	0	Demeter, Hestia, Hera, Hades

A combination of counterfactual responses is called a **response pattern** or **response type**.

3.2 Two Treatments: Sixteen Response Types

With two dichotomous treatments each individual has four counterfactual outcomes, so there are $2^4 = 16$ response types (Miettinen 1982).

Table 2: Response types for two dichotomous treatments; superscripts are (a, e) (Hernán and Robins 2020, Table 5.2, p. 65)

Type	$Y^{1,1}$	$Y^{0,1}$	$Y^{1,0}$	$Y^{0,0}$
1	1	1	1	1
2	1	1	1	0
3	1	1	0	1
4	1	1	0	0
5	1	0	1	1
6	1	0	1	0
7	1	0	0	1
8	1	0	0	0
9	0	1	1	1
10	0	1	1	0
11	0	1	0	1
12	0	1	0	0
13	0	0	1	1
14	0	0	1	0
15	0	0	0	1
16	0	0	0	0

3.3 Types Without Interaction

In six types the effect of each treatment does not depend on the other:

- type 1 (dies under every joint treatment) and type 16 (survives under every joint treatment);
- type 4: dies only if given vitamins (E alone matters);
- type 13: dies only if not given vitamins;
- type 6: dies only if transplanted (A alone matters);
- type 11: dies only if not transplanted.

If everyone has one of types 1, 4, 6, 11, 13, 16, there is no interaction between A and E on the additive scale.

If everyone were of type 1 or 16, the sharp causal null hypothesis would hold for the joint treatment (A, E) (Hernán and Robins 2020, 65).

3.4 Additive Interaction Requires Interaction Types

Additive interaction implies that some individuals belong to at least one of three classes (Greenland and Poole 1988), each invariant to recoding A and E:

1. outcome under only **one** of the four joint treatments: types 8, 12, 14, 15;
2. outcome under **two** joint treatments, with the effect of each treatment reversed across levels of the other: types 7, 10;
3. outcome under **three** of the four joint treatments: types 2, 3, 5, 9.

No additive interaction implies either that nobody is in these classes or that equal deviations of opposite sign cancel exactly (e.g., equal proportions of types 7 and 10, or of types 8 and 12).

Example 3.1 (Cancellation of Interaction Types)**.** Let a fraction q of the population be type 8, a fraction q be type 12, and the rest type 16. Reading the risks off Table 2: $p_{11} = q$ (type 8), $p_{01} = q$ (type 12), $p_{10} = 0$, $p_{00} = 0$. Then

$$\begin{aligned} p_{11} - p_{01} - p_{10} + p_{00} &= q - q - 0 + 0 \\ &= 0, \end{aligned}$$

so there is no interaction on the additive scale in the population, even though every type-8 and type-12 individual has an effect of A that depends on E.

Miettinen's 16 types are not invariant to switching the labels "0" and "1"; Greenland and Poole's three classes are (Hernán and Robins 2020, 65, margin note). Greenland, Lash, and Rothman (2008) discuss such cancellations further.

Technical Point 5.2: Monotonicity

For one treatment, $Y^{a=0} > Y^{a=1}$ only for the "helped". If nobody is helped, every individual has $Y^{a=1} \geq Y^{a=0}$, and the causal effect of A on Y is **monotonic**.
For two treatments, the effects of A and E are monotonic if every $Y^{a,e}$ is nondecreasing in both a and e, i.e., nobody has

- $Y^{a=1,e=1} = 0$ and $Y^{a=0,e=1} = 1$;
- $Y^{a=1,e=1} = 0$ and $Y^{a=1,e=0} = 1$;
- $Y^{a=1,e=0} = 0$ and $Y^{a=0,e=0} = 1$;
- $Y^{a=0,e=1} = 0$ and $Y^{a=0,e=0} = 1$.

Source: Hernán and Robins (2020, Technical Point 5.2, p. 66).

Fine Point 5.1: Detecting Synergistic Types

Do individuals exist who develop the outcome under both treatments but under neither alone, i.e., with $Y^{a=1,e=1} = 1$ and $Y^{a=0,e=1} = Y^{a=1,e=0} = 0$ (types 7 and 8)? A **sufficient** condition (VanderWeele and Robins 2007a, 2008) is

$$p_{11} - p_{01} - p_{10} > 0, \quad \text{equivalently} \quad p_{11} - p_{01} > p_{10}.$$

If the effects are monotonic, a weaker sufficient condition is superadditive interaction:

$$p_{11} - p_{01} > p_{10} - p_{00},$$

which then implies type 8 exists (monotonicity rules out type 7).
All three risks can be computed in a randomized experiment on A and E, so the existence of types 7 and 8 can be checked empirically (Hernán and Robins 2020, Fine Point 5.1, p. 67). Because the first condition is sufficient but not necessary, it can fail even when types 7 and 8 exist, and it is strong enough to miss most such cases. The monotonic-case result was reported by Greenland and Rothman and appears in Greenland, Lash, and Rothman (2008).

In genetics, the existence of type-8 individuals is called **compositional epistasis**; VanderWeele (2010a) reviews tests for it.

4 5.4 Sufficient Causes (pp. 66-69)

The variety of response types shows that A is not the only determinant of Y. The sufficient-component-cause framework represents the other determinants as **background factors**.

Definition 4.1 (Minimal Sufficient Cause)**.** A **sufficient cause** of Y is a set of conditions that together inevitably produce Y. It is **minimal** if it contains the smallest set of background factors that, with the treatment component, suffices. Its elements are its **component causes**; "sufficient-component causes" refers to both.

By definition, the dichotomous background factors U cannot be intervened on and cannot be affected by treatment A (Hernán and Robins 2020, 67, margin note).

4.1 One Treatment: Three Sufficient Causes

In the oversimplified heart transplant example (Hernán and Robins 2020, 66–67):

Sufficient cause	Treatment component	Background factor (example)
$A = 1$ with $U_1 = 1$	transplant	allergy to anesthesia
$A = 0$ with $U_2 = 1$	no transplant	ejection fraction below 20%
$U_0 = 1$	none	pancreatic cancer at study start

Figure 5.1 of the book draws each sufficient cause as a circle ("causal pie") divided into its components.

4.2 Effect Size Depends on Background Factors

Compare two populations, identical except that $U_1 = 1$ (allergy) has prevalence

- 1% in the first;
- 10% in the second.

Randomize half of each population to $A = 1$. The average causal effect of transplant on death is greater in the second population, because the sufficient cause "$A = 1$ plus $U_1 = 1$" is ten times more common there.

This is the Chapter 4 point that the magnitude of a causal effect depends on the distribution of effect modifiers, made visible with sufficient-component causes (Hernán and Robins 2020, 68).

4.3 Two Treatments: Nine Sufficient Causes

With treatments A and E there are 9 possible sufficient causes (Greenland and Poole 1988), with treatment components

1. $A = 1$ only;
2. $A = 0$ only;
3. $E = 1$ only;
4. $E = 0$ only;
5. $A = 1$ and $E = 1$;

6. $A = 1$ and $E = 0$;
7. $A = 0$ and $E = 1$;
8. $A = 0$ and $E = 0$;
9. neither A nor E.

Each also contains background factors from $U_1, \ldots, U_8$ and U_0 (Figure 5.2 of the book).

Not all 9 need exist. If vitamins $E = 1$ never kill anyone, whatever A is, then the 3 sufficient causes with component $E = 1$ are absent; their existence would mean some individuals (e.g., those with $U_3 = 1$) would be killed by vitamins, i.e., saved by withholding them (Hernán and Robins 2020, 69).

5 5.5 Sufficient Cause Interaction (pp. 69-71)

The counterfactual definition of interaction (Definition 1.2) is a contrast of counterfactual risks. It can be identified in an ideal randomized experiment on A and E without any knowledge of the mechanisms by which the treatments act.

A second concept of interaction refers to mechanisms directly.

Definition 5.1 (Sufficient Cause Interaction)**.** "A sufficient cause interaction between A and E exists in the population if A and E occur together in a sufficient cause" (Hernán and Robins 2020, 69).

Example 5.1 (Background Factor U_5)**.** Suppose individuals with $U_5 = 1$ develop the outcome when receiving both vitamins and transplant but not when receiving only one of them. Then a sufficient cause interaction exists if anyone has $U_5 = 1$. Hence, if some individual has $Y^{a=1,e=1} = 1$ and $Y^{a=0,e=1} = Y^{a=1,e=0} = 0$, a sufficient cause interaction between A and E is present (Hernán and Robins 2020, 69–70).

5.1 Synergism and Antagonism

- **Synergism**: $A = 1$ and $E = 1$ are components of the same sufficient cause.
- **Antagonism**: $A = 1$ and $E = 0$ (or $A = 0$ and $E = 1$) are components of the same sufficient cause.

Antagonism between A and E can be viewed as synergism between A and "no E" (or between "no A" and E).

Rothman (1976) described synergism and antagonism within the sufficient-component-cause framework (Hernán and Robins 2020, 71, margin note).

5.2 Detecting Synergism Without Knowing the Mechanisms

Sufficient cause interaction is defined through mechanisms, yet sometimes it can be detected with no knowledge of them: if the inequalities of Fine Point 5.1 hold, synergism between A and E exists.

This is not surprising, for two reasons (Hernán and Robins 2020, 71): response types correspond to sufficient causes (Fine Point 5.2), and the inequalities are sufficient but not necessary, so they can fail even when synergism exists.

Fine Point 5.2: From Response Types to Component Causes

Table 3: Mapping between response types and sufficient-component causes for one treatment (Hernán and Robins 2020, Fine Point 5.2, p. 70)

Type	$Y^{a=0}$	$Y^{a=1}$	Component causes
Doomed	1	1	$U_0 = 1$ or $\{U_1 = 1$ and $U_2 = 1\}$
Helped	1	0	$U_0 = 0, U_1 = 0, U_2 = 1$
Hurt	0	1	$U_0 = 0, U_1 = 1, U_2 = 0$
Immune	0	0	$U_0 = 0, U_1 = 0, U_2 = 0$

Each combination of component causes gives exactly one response type, but a response type can arise from several combinations (e.g., "doomed" arises from any combination with $U_0 = 1$, or with $U_1 = 1$ and $U_2 = 1$).

Exchangeability in terms of component causes (Hernán and Robins 2020, Fine Point 5.2, p. 70). For a dichotomous treatment and outcome, $Y^a \perp\!\!\!\perp A$ means $\Pr[Y^{a=1} = 1 \mid A = 1] = \Pr[Y^{a=1} = 1 \mid A = 0]$ and $\Pr[Y^{a=0} = 1 \mid A = 1] = \Pr[Y^{a=0} = 1 \mid A = 0]$.

- Those with $Y^{a=1} = 1$ are the doomed and the hurt, i.e., those with $U_0 = 1$ or $U_1 = 1$.
- Those with $Y^{a=0} = 1$ are the doomed and the helped, i.e., those with $U_0 = 1$ or $U_2 = 1$.

Substituting these sets, exchangeability holds when $\Pr[U_0 = 1 \text{ or } U_1 = 1 \mid A = 1] = \Pr[U_0 = 1 \text{ or } U_1 = 1 \mid A = 0]$ and $\Pr[U_0 = 1 \text{ or } U_2 = 1 \mid A = 1] = \Pr[U_0 = 1 \text{ or } U_2 = 1 \mid A = 0]$.

See Greenland and Brumback (2002), Flanders (2006), and VanderWeele and Hernán (2006); VanderWeele and Robins (2008) generalized some results to two or more treatments.

Fine Point 5.3: "Biologic" Interaction

Sufficient cause interaction is often called **biologic interaction** (Rothman et al. 1980), but it need not involve treatments acting on each other.

Example 5.2 (Two Alleles (Hernán and Robins 2020, Fine Point 5.3, p. 71))**.** Let A and E indicate a deleterious mutation in each of the two alleles of a gene that produces an essential protein (VanderWeele and Robins 2007a). Infants with both mutations ($A = 1$, $E = 1$) lack the protein and die within a week; those with one or no mutation survive. There is synergism, since a sufficient cause of death contains $A = 1$ and $E = 1$, yet the alleles arguably do not physically act on each other.

6 5.6 Counterfactuals or Sufficient-Component Causes? (pp. 71-74)

The two frameworks answer different questions:

	Sufficient-component causes	Counterfactuals
Starts from	a particular effect	a particular cause or intervention
Asks	"how does it happen?"	"what happens?"
Describes mechanisms	yes	no

For estimating average causal effects of hypothetical interventions, the subject of the book, the counterfactual framework is the natural one.

The sufficient-component-cause model asks which events might have caused a given effect; the counterfactual model asks what would have happened had a factor been set to a different level than it was (Hernán and Robins 2020, 71). In philosophy the sufficient-component-cause framework goes back to Mackie (1965), whose **INUS condition** for Y is an **I**nsufficient but **N**ecessary part of a condition which is itself **U**nnecessary but exclusively **S**ufficient for Y. A counterfactual framework of causation was already hinted at by Hume (1748).

6.1 Strengths and Limitations of Sufficient-Component Causes

Useful for teaching, because they illustrate

- why the size of a causal effect depends on the distribution of background factors (effect modifiers);
- how effect modification, interaction, and synergism relate.

Limited for data analysis, because in its classical form the framework

- is deterministic;
- gives conclusions that depend on the coding of the outcome;
- is restricted to dichotomous treatments and outcomes;
- requires large amounts of data to study the fine distinctions it makes.

Extensions to stochastic settings and to categorical or ordinal treatments might widen its use: VanderWeele (2010b) extended it to 3-level treatments, and VanderWeele and Robins (2012) related stochastic counterfactuals to stochastic sufficient causes (Hernán and Robins 2020, 72). The counterfactual framework will likely remain the one most often used, and apparent alternatives such as causal diagrams and decision theory are essentially equivalent to it (Chapter 6).

Fine Point 5.4: Attributable Fractions Do Not Add

Suppose the excess fraction (Fine Point 3.5) is 75% for A and 50% for E. A joint intervention cannot prevent $75\% + 50\% = 125\%$ of cases: no intervention, single or joint, can prevent more than 100%.
The resolution: an individual like Zeus with $U_5 = 1$, treated with $A = 1$ and $E = 1$, would not have been a case had **either** treatment been withheld, so Zeus is counted in both the 75% and the 50%.
When A and E can be components of the same sufficient cause, the fraction of disease attributable to each separately makes little sense (Hernán and Robins 2020, Fine Point 5.4, p. 72). Phenylketonuria example: mental retardation occurs in genetically susceptible individuals who eat certain foods. Removing the foods would prevent all cases, and so would replacing the susceptibility genes, so the cases are both "100% environmental" and "100% genetic". See Rothman, Greenland, and Lash (2008).

Technical Point 5.3: Monotonicity Rules Out Some Sufficient Causes

If smoking ($A = 1$) never prevents heart disease and physical inactivity ($E = 1$) never prevents heart disease, then no sufficient cause can contain $A = 0$ or $E = 0$.
If a sufficient cause containing $A = 0$ existed, some individuals (e.g., those with $U_2 = 1$) would develop the outcome when unexposed, i.e., treating them ($A = 1$) would prevent their outcome, contradicting monotonicity; the same argument applies to $E = 0$. Figure 5.3 of the book crosses out the sufficient causes excluded this way (Hernán and Robins 2020, Technical Point 5.3, pp. 73-74).

7 Summary

- **Interaction** between two treatments A and E is defined by joint counterfactuals $Y^{a,e}$: on the additive scale, $\Pr[Y^{1,1} = 1] - \Pr[Y^{0,1} = 1] \neq \Pr[Y^{1,0} = 1] - \Pr[Y^{0,0} = 1]$, a definition symmetric in A and E; a multiplicative version uses risk ratios.
- **Identification** requires exchangeability, positivity, and consistency for the joint treatment (A, E); if E is randomized, interaction coincides with effect modification by E.
- **Effect modification** can occur without interaction (surrogate modifiers).
- With deterministic, dichotomous counterfactuals there are **16 response types**; additive interaction requires individuals of interaction types, but their presence can cancel out.
- **Sufficient cause interaction** means A and E appear in the same sufficient cause; synergism can sometimes be detected from counterfactual risks alone (Fine Point 5.1).
- The counterfactual framework answers "what happens?" and is the one used in the rest of the book; sufficient-component causes answer "how does it happen?" and are mainly a teaching tool.

Looking ahead: Chapter 6 introduces causal diagrams, which the book describes as essentially equivalent to the counterfactual framework.

8 References

Hernán, Miguel A, and James M Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC. https://miguelhernan.org/whatifbook.